F.A.R. PART 23 SUB-PART D

Design & Construction

Part 23 Sub-Para No.	Requirement	Compliance		Reference
23.689 (e)	Cable Systems - Attachment of Turnbuckles to Parts Subject to Angular Motion.	Yes	All turnbuckles are positioned away from levers.	
23.689 (f)	Cable Systems - Tab Control Cables.	Yes	Cables used in the secondary control circuit are not less than $\frac{1}{8}$" diameter.	
23.693	Joints - Special Factors of Safety.	Yes	These joints have been stressed or tested satisfactorily.	REPORT ANZ AT-014
23.697 (a)	Wing Flap Control - Ability to Remain in a Set Position.	Yes	A positive lock is incorporated in the flap handle.	
23.697 (b)	Wing Flap Control - Rate of Movement to give Satisfactory Characteristics.	Yes	Rate of movement for electric or manual flap control is adequate at Vf and below.	Flight test. Report guide P69 (FIRG)
23.699 (a)(1)(2) (b)(1)(2)	Wing Flap Position Indication.	Yes	Flap position indicator fitted.	
23.701 (a)	Flap Interconnection - Synchronisation.	Yes	Mechanical interconnection (push-pull system) fitted.	

F.A.R. PART 23 SUB-PART D

Design & Construction

Part 23 Sub-Para No.	Requirement	Compliance		Reference
23.701 (b)	Flap Interconnection - Unsymmetrical Loads.	N/A	The Airtourer is a single-engined aircraft.	
23.723 (a)	Shock Absorption Tests - Tests to Prove Landing Limit Load Factors.	Yes		STRUCTURAL TEST REPORTS ANZ AT-007 & AT-008.
23.723 (b)	Shock Absorption Tests - Strength of Landing Gear for Reserve Energy Absorption Condition.	Yes		STRUCTURAL TEST REPORTS ANZ AT-007 & AT-008.
23.725 (a)(b)(c)(d)(e)(f)	Landing Gear - Limit Drop Tests.	Yes	The conditions of this requirement have been complied with during drop testing.	STRUCTURAL TEST REPORTS ANZ AT-007, AT-008, AT-010 & AT-012.
23.726	Ground Load Dynamic Tests	N/A	Compliance with load requirements of 23.479 through 23.483 were not shown by drop testing.	
23.727 (a)(b)	Reserve Energy Absorption Drop Test	Yes		STRUCTURAL TEST REPORTS ANZ AT-007, AT-008, AT-010 & AT-012.
23.729	Retracting Mechanism.	N/A	The Airtourer has a fixed undercarriage.	

F.A.R. PART 23 SUB-PART D

Design & Construction

Part 23 Sub-Para No.	Requirement	Compliance		Reference
23.731 (a)	Wheels - Approval.	Yes	Each main and nose wheel is approved.	
23.731 (b)	Wheel Static Load Rating		Nose Wheel Static load = 453 lbs Static load rating > 900 lbs Main Wheel Static load = 825 lbs Static load rating > 900 lbs	STRUCTURAL REPORTS ANZ AT-007 & AT-012 DUNLOP AIRCRAFT TYRE MANUAL
23.731 (c)	Wheel Limit Load Rating		Main wheel maximum limit radial load = 2770 lbs. Nose wheel maximum limit radial load = 2190 lbs. Wheel limit load rating > 3000 lbs	STRUCTURAL REPORTS ANZ AT-007 & AT-012
23.733 (a)(1)	Tyres - Main Wheel Rating	Yes	The max. static load/wheel = 825 lbs. The tyre rating (Dunlop 5.00 x 5) at 30 p.s.i. = 900 lbs.	
23.733 (a)(2)	Tyres - Nose Wheel Rating	Yes	With 1g down and .33g forward in the most adverse c.g. position the load is 820 lbs. The static load of the 5.00 x 5 Dunlop tyre at 30 p.s.i. is 900 lbs.	

F.A.R. PART 23 SUB-PART D

Design & Construction

Part 23 Sub-Para No.	Requirement	Compliance		Reference
23.735 (a)	Brakes - Kinetic Energy Capacity	Yes	Kinetic energy requirement $= \frac{0.0444WVso^2}{N} = 105,500$ ft.lb for W = 1900 lb and stall speed = 50 knots. The capacity of the brakes is 130,000 ft.lbs.	VICTA REPORT GEN-107
23.735 (b)	Brakes - Ability to Prevent Rolling.	Yes	The brakes will hold the aircraft on a paved runway with take-off power on the engine.	FIRG Pg 60
23.737 through 23.757	Skis, Floats & Hulls	N/A	The Airtourer is not equipped for ski or water operation.	
23.771	Pilot Compartment	Yes		
23.773 (a) (b)	Pilot Compartment View	Yes	Complies for both day and night flying	FIRG P22.
23.775 (a)	Windshields & Windows	N/A	There are no internal glass panels on the Airtourer.	
23.775 (b)(c)	Windshields & Windows - Pressurised Aircraft.	N/A	The Airtourer is unpressurised.	

F.A.R. PART 23 SUB-PART D

Design & Construction

Part 23 Sub-Para No.	Requirement	Compliance		Reference
23.775 (d)	Windshields & Windows - Luminous Transmittance.	Yes	The specified windshield and side windows are manufactured from transparent "Perspex" acrylic resin sheet, the luminous transmittance value of which is 92%.	
23.775 (e)	Windshields & Windows - Operation above 25,000 ft.	N/A	The Airtourer is not certified to operate above 25,000 ft.	
23.777 (a)	Cockpit Controls - Location & Identification.	Yes		
23.777 (b)	Cockpit Controls - Unrestricted Movements.	Yes		
23.777 (c)	Cockpit Controls - Location of Power Plant Controls.	N/A	The Airtourer has only one engine.	
23.777 (d)	Cockpit Controls - Location of Wing Flap Control	No	Electric flap control in correct position if employed but manual flap lever is incorrectly placed.	
23.777 (e)	Cockpit Controls - Location of Landing Gear Control.	N/A	The Airtourer has no landing gear control.	

Design & Construction

Part 23 Sub-Para No.	Requirement	Compliance		Reference
23.777 (f)	Cockpit Controls - Location of Fuel Feed Selector.	Yes		
23.779	Motion & Effect of Cockpit Controls.	Yes		
23.781	Cockpit Control Knob Shape	No	The manual flap control is a rod with a release button on the end. The electric flap knob also does not conform with the required shape.	
23.783 (a)(b)	Doors	N/A	No passenger accommodation is provided on the Airtourer.	
23.785 (a)	Seats & Berths - Strength	Yes	The seats have been designed to the ANO requirements which satisfy 23.561.	TR-02 (p145)
23.785 (b)	Approval of Seats	Yes	Seats are approved as part of the overall aircraft design. They are not bought out items.	
23.785 (c)	Pilot Seat - Loads due to Control Forces	Yes	In no case will the pilot forces to the primary flight controls produce a critical load on the pilot seat.	

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Design & Construction

Part 23 Sub-Para No.	Requirement	Compliance		Reference
23.785 (d)	Seats - Suitability to Wear Parachutes.	Yes	The seats are suitable for the wearing of parachutes.	
23.785 (e)	Seats & Berths - Design of Berths.	N/A	There are no berths fitted to the Airtourer.	
23.785 (f)	Seats & Berths - Proof of Compliance.	Yes	The seats have been stressed for the design loads.	VICTA REPORT TR-02.
23.785 (g)	Seats & Berths - Protection of Occupants.	Yes	A safety belt and shoulder harness protects the head from injury.	
23.787 (a)	Cargo Compartments - Placards	Yes	Placards are provided as required.	
23.787 (b)	Cargo Compartments - Avoidance of Hazards.	Yes	Tie-down points are provided for securing baggage.	
23.787 (c)	Cargo Compartments - Protection of Passengers from Injury.	Yes	The seat back will react the load provided it is tied down securely.	
23.807 (a)	Emergency Exits - Number & Location.	Yes	The Airtourer has the required number of emergency exits; one sliding canopy for pilot and passenger.	

F.A.R. PART 23 SUB-PART D

Design & Construction

Part 23 Sub-Para No.	Requirement	Compliance		Reference
23.807 (b)	Emergency Exits - Type & Operation.	No	The canopy cannot be opened by the occupants at V_D.	
23.807 (c)	Emergency Exits - Functioning Tests.	No	Refer 23.807(b).	
23.831	Ventilation.	Yes	Tests carried out satisfactorily.	FIRG P88
23.841 & 23.843	Pressurisation	N/A	The Airtourer is not pressurised.	
23.853 (a)	Fire Protection - Compartment Interiors - Flash Resistant Materials.	Yes	Materials are all flame resistant permitting smoking in flight	Type inspection report P18 (T.I.R.)
23.853 (b)	Fire Protection - Smoking.	Yes	See 23853(a)	
23.853 (c)	Placarding for Non-Smoking.	N/A	Smoking is not allowed.	
23.859	Combustion Heater Fire Protection.	N/A	No combustion heater is provided.	

F.A.R. PART 23 SUB-PART D

Design & Construction

Part 23 Sub-Para No.	Requirement	Compliance		Reference
23.867	Lightning Evaluation	Yes	With the all-metal construction of the Airtourer the effect of a lightning strike is likely to be minimal.	
23.871	Levelling Means	Yes	Defined as canopy rails.	

F.A.R. PART 23 SUB-PART E

Power Plant

Part 23 Sub-Para No.	Requirement	Compliance		Reference
23.901	Installation	Yes		
23.903	Engines - Type Certification	Yes	Engine is type certificated	Type Certificate E274
23.905 (a)	Propellors - Type Certification	Yes	Propellor is type certificated.	Type Certificate P-920-8
23.905 (b)	Propellors - Engine power and Rotational Speed Limitations.	Yes	Propellor/engine combination is approved.	P-920-8
23.905 (c)	Propellors - Unfeathering Means.	NA	Propellors are non-feathering.	
23.907	Propellor Vibration	Yes	Propellor/engine combination is approved.	P-920-8

F.A.R. PART 23 SUB-PART E

Power Plant

Part 23 Sub-Para No.	Requirement	Compliance		Reference
23.909	Turbosuperchargers	N/A	No turbosuperchargers are fitted to the engine.	
23.925 (a)	Propellor - Ground Clearance	Yes		Appendix to ground inspection
23.925 (b)	Propellor - Water Clearance	N/A	The Airtourer does not operate on water.	
23.925 (c)(1)(2)(3)	Propellor - Structural Clearance	Yes		
23.933 through 23.939	Reversing Systems, Turbo-propellor - Drag Limiting Systems, Turbine Engine Power Plant Operating Characteristics.	N/A	None of these systems is incorporated in the Airtourer.	
23.951 (a)	Fuel System - General	Yes		
23.951 (b)	Fuel System - Drawing from one Tank at a Time.	N/A	The Airtourer has only one fuel tank.	
23.953	Fuel System - Independence	N/A	The Airtourer has only one engine.	

F.A.R. PART 23 SUB-PART E

Power Plant

Part 23 Sub-Para No.	Requirement	Compliance		Reference
23.954	Fuel System - Lightning Protection.	Yes		
23.955 (a)	Fuel Flow - General	Yes		
23.955 (b)	Fuel Flow - Gravity Systems	N/A	The Airtourer does not have a gravity fuel system.	
23.955 (c)	Fuel Flow - Pump Systems	Yes		? measurements ?
23.955 (d)(1)	Fuel Flow - Fuel Transfer Systems.	N/A	No transfer system on Airtourer	
23.955 (d)(2)	Fuel Flow - Auxiliary Tanks	N/A	This requirement applies only to aircraft having small auxiliary tanks feeding into a large main tank.	
23.955 (e)	Multiple Fuel Tanks	N/A	The Airtourer has only one tank.	
23.955 (f)	Turbine Engine Fuel Systems	N/A	The Airtourer does not have a turbine engine.	

F.A.R. PART 23 SUB-PART E

Power Plant

Part 23 Sub-Para No.	Requirement	Compliance		Reference
23.957	Fuel Flow - Flow Between Interconnected Tanks.	N/A	This requirement applies only to aircraft having gravity fuel feed systems.	
23.959	Unusable Fuel Supply	Yes		FIRG P75-76
23.961	Fuel System Hot Weather Operation.	Yes		FIRG P76
23.963 (a)	Fuel Tanks - Strength	Yes		
23.963 (b)	Fuel Tanks - Flexible	Yes		
23.963 (c)	Fuel Tanks - Integral Facilities for Interior Inspection & Repair	N/A	Integral construction is not used.	
23.963 (d)	Fuel Tanks - Usable Capacity	Yes	The Airtourer has a fuel capacity sufficient for over 2½ hrs at maximum continuous power.	
23.963 (e)	Calibration of fuel quantity indicator	Yes	Fuel quantity indicator complies	

F.A.R. PART 23 SUB-PART E

Power Plant

Part 23 Sub-Para No.	Requirement	Compliance		Reference
23.965 (a)(1)	Fuel Tank Tests - Tanks not Supported by Aircraft Structure.	N/A	The Airtourer does not use fuel tanks of this type.	
23.965 (a)(2)	Fuel Tank Tests - Integral Tank Pressures.	N/A	Integral tank construction is not employed.	
23.965 (a)(3)	Fuel Tank Tests - Non-metallic Tanks	Yes	The tank has been satisfactorily tested to a pressure of 2 p.s.i.	REPORT ANZ AT-016
23.965 (b)(1)(2)(3)(4)(5)	Fuel Tanks - Vibration Testing	Yes	This requirement can be ignored because of the flexible tank design on the Airtourer.	
23.965 (c)	Fuel Tanks - Sealing & Construction.	N/A	Integral tanks are not used.	
23.965 (d)	Fuel Tank Tests - Tanks with Non-metallic Liners	Yes	Although this test has not been carried out compliance is assumed by the original design and the long term service history of the tanks.	
23.967 (a)(1)(2)(3)	Fuel Tank Installation	Yes		

F.A.R. PART 23 SUB-PART E

Power Plant

Part 23 Sub-Para No.	Requirement	Compliance		Reference
23.967 (a)(4)	Fuel Tank Installation - Projections.	Yes	All protuberances in the tank bay are covered with 'Scotch cal' pressure sensitive tape.	
23.967 (a)(5)	Fuel Tank Installation - Maintenance of Positive Pressure	Yes		
23.967 (a)(6)	Fuel Tank Installation - Syphoning.	Yes	No syphoning will occur with the filler arrangement on the Airtourer.	
23.967 (b)(c)	Fuel Tank Installation	Yes	Tank bay is adequately vented and drained. No tank fwd of firewall	
23.967 (d)	Fuel Tank Installation - Tank in Personnel Compartment.	Yes	There is a double skin thickness between the fuel tank and personnel compartment.	
23.969	Fuel Tank Expansion Space	Yes	Vent discharges clear of aircraft	
23.971	Fuel Tank Sump	Yes	The fuel tank sump has a capacity of .11 gal or .38% of the total capacity.	
23.973 (a)	Fuel Tank Filler Connection - Marking.	Yes	Placard provided.	

F.A.R. PART 23 SUB-PART E

Power Plant

Part 23 Sub-Para No.	Requirement	Compliance		Reference
23.973 (b)	Fuel Tank Filler Connection - Spilled Fuel.	Yes		
23.973 (c)	Fuel Tank Filler Connection - Filler Cap Sealing.	Yes		
23.975 (a)(1)	Fuel Tank Vents - Method of Venting Tanks.	Yes	Each tank is vented from the top and each vent outlet is located to minimise the possibility of its being obstructed by ice or other foreign matter.	
23.975 (a)(2)	Fuel Tank Vents - Prevention of Syphoning.	Yes	The position of the vent outlet and inlet in relation to the fuel tank will prevent any syphoning.	
23.975 (a)(3)	Fuel Tank Vents - Rapid Relief of Pressure Difference.	Yes	Venting adequate.	
23.975 (a)(4)	Fuel Tank Vents - Interconnection of Air Spaces.	N/A	The Airtourer has only one tank.	
23.975 (a)(5)	Fuel Tank Vents - Ability to Drain.	Yes	Moisture cannot accumulate in the Airtourer fuel tank venting system.	

F.A.R. PART 23 SUB-PART E

Power Plant

Part 23 Sub-Para No.	Requirement	Compliance		Reference
23.975 (a)(6)	Fuel Tank Vents - Discharge of Vented Fuel.	Yes	The fuel vent is at the rear of the aircraft.	
23.975 (b)	Carburettor Vapour Vents	N/A	The carburettors on the Airtourer do not require vapour vents.	
23.975 (c)	Fuel Tank Vents - Acrobatic Category Aircraft.	Yes	Bulkhead	
23.977	Fuel Tank Outlet	Yes	The diameter of the strainer is greater than that of the outlet.	
23.991 (a)	Main Fuel Pumps	Yes		
23.991 (b)	Emergency Pump	Yes	The emergency pump is electrically operated.	
23.995 (c)	Fuel Pumps - Warning Means	N/A	Main and emergency pumps are used together only on take-off and landing.	
23.993	Fuel System Lines & Fittings.	Yes		

F.A.R. PART 23 SUB-PART E

Power Plant

Part 23 Sub-Para No.	Requirement	Compliance		Reference
23.994	Fuel System Components	Yes		
23.995	Fuel Valves & Controls	Yes		
23.997	Fuel Strainer or Filter.	Yes		
23.999	Fuel System Drains	Yes		
23.1001	Fuel Jettisoning System	N/A	The take-off and landing weights are the same and no jettisoning system is employed.	
23.1011 (a)	Oil System - General	Yes	The engine is of the wet sump type and is equipped with an oil cooler.	
23.1011 (b)(c)	Oil System - Usable Capacity.	Yes	Oil reservoir capacity is adequate.	
23.1011 (d)	Oil System - Oil Transfer Pump	N/A	There is no oil transfer pump in the Airtourer oil system.	
23.1013 (a)(1)(2) (b)(1)(2)	Oil Tanks	Yes	The oil tanks are integral with the engine.	

F.A.R. PART 23 SUB-PART E

Power Plant

Part 23 Sub-Para No.	Requirement	Compliance		Reference
23.1013 (c)	Oil Tank Filler Connection.	Yes	Placard provided.	
23.1013 (d)(e)	Oil Tank Vent & Outlet	Yes	Aerobatic P.	
23.1013 (f)	Oil Tanks - Flexible Liners	N/A	There are no flexible liners used in the Airtourer oil tank.	
23.1015 (a)(b)	Oil Tank Tests	N/A	The Airtourer oil tank is part of the certified engine.	
23.1017 (a)	Oil Lines & Fittings - General	Yes	Inside diameters of engine oil inlet and outlet pipes are the same as the inlet and outlet connections on the engine.	
23.1017 (b)	Oil Breather Lines	Yes		
23.1019	Oil Strainer or Filter.	Yes	The oil strainer or filter is part of the certificated engine.	
23.1021 (a)(b)	Oil System Drains - Drainage of Entire System & Means of Locking.	Yes	These items are part of the certificated engine. They are wire locked.	

F.A.R. PART 23 SUB-PART E

Power Plant

Part 23 Sub-Para No.	Requirement	Compliance		Reference
23.1023	Oil Radiators - Suitability to Withstand Vibration, Inertia & Oil Pressure Loads.	Yes	The oil radiators meet the requirements load - down in the engine manufacturers specification.	
23.1027	Propeller Feathering System - Retention of Sufficient Oil for Feathering & Functioning of Feathering System on Retained Oil	N/A	No propellor feathering system is fitted.	
23.1041 through 23.1047	Cooling	Yes	Cooling requirements met.	FIRG P42, P80-84
23.1061 & 23.1063	Liquid Cooling	N/A	The Airtourer engine is air cooled.	
23.1091 (a)(b)(c)(d)(e)	Induction System - Air Induction	Yes	A normal and an alternate air intake are provided and these meet the requirements.	
23.1093	Induction System - Icing Protection.	Yes	Carburettor air heat rise meets requirements	FIRG P77-78

F.A.R. PART 23 SUB-PART E

Power Plant

Part 23 Sub-Para No.	Requirement	Compliance		Reference
23.1095 23.1097 23.1099	Carburettor Fluid De-icing System	N/A	The Airtourer engine is not equipped with a fluid system of carburettor de-icing.	
23.1101	Carburettor Air Preheater Design	Yes		
23.1103	Induction System Ducts	Yes		
23.1105	Induction System Screens	Yes		
23.1111	Turbine Engine Bleed Air System	N/A	The Airtourer is not fitted with a turbine engine.	
23.1121	Exhaust System - General	Yes		TIR P22
23.1123	Exhaust Manifold	Yes		
23.1125	Exhaust Heat Exchangers	Yes		
23.1141	Power Plant Controls - General	Yes		

F.A.R. PART 23 SUB-PART E

Power Plant

Part 23 Sub-Para No.	Requirement	Compliance		Reference
23.1143	Engine Power & Thrust, & Supercharger Controls	Yes	With only one engine and no supercharger, only one power control need be provided.	
23.1145	Ignition Switches	Yes	Only one ignition switch is provided for the single engine.	
23.1147	Mixture Controls	Yes	The mixture control is arranged to prevent confusion with other controls.	
23.1149	Propeller Speed & Pitch Controls	N/A	Only one control is required.	
23.1153	Propeller Feathering Controls	N/A	No feathering control is provided.	
23.1155	Turbine Engine Reverse Thrust	N/A		
23.1157	Carburettor Air Temperature Control	Yes		
23.1163	Power Plant Accessories	Yes		
23.1165 (a)(b) (c)(1)(2)(3) (d)	Engine Ignition System - Battery Ignition Systems.	N/A	These requirements apply only to battery ignition systems.	

F.A.R. PART 23 SUB-PART E

Power Plant

Part 23 Sub-Para No.	Requirement	Compliance		Reference
23.1183 (a)(b)	Power Plant Fire Protection - Lines & Fittings	Yes	All the appropriate lines and fittings on the Airtourer are fire resistant.	
23.1189	Power Plant Fire Protection - Shut off Means.	N/A	This requirement applies to multi-engine and turbine engine installations.	
23.1191	Firewalls	Yes	All openings are sealed with fireproof fittings, and construction is of .028" mild steel sheet.	
23.1193 (a)(b)(c)(d)(e)	Cowlings - Design Requirements	Yes	Compliance with 23.1193(d) is based upon aluminium alloy in dimensions appropriate to its purpose and the definition of "Fire Resistant" in FAR Part 1, "Definitions & Abbreviations".	

F.A.R. PART 23 SUB-PART F

Equipment

Part 23 Sub-Para No.	Requirement	Compliance		Reference
23.1301	Equipment - Function & Installation.	Yes		
23.1303	Flight & Navigation Instruments	Yes	The minimum instruments are provided.	
23.1305	Power Plant Instruments.	Yes	The minimum instruments are provided.	
23.1307	Miscellaneous Equipment	Yes		
23.1321	Instrument Arrangement & Visibility.	Yes		
23.1323	Airspeed Indicating System	Yes	Installation error is within limits.	FIRG P89
23.1325 (a)(b)(1)	Static Pressure System	Yes		
23.1325 (b)(2)	Static Pressure System Test	Yes	A satisfactory leak test was carried out by M.O.T./CAD	
23.1325 (b)(3)	Static Pressure System - Icing	No	No special precaution has been taken to prevent icing of the static pressure port.	[illegible]

F.A.R. PART 23 SUB-PART F

Equipment

Part 23 Sub-Para No.	Requirement	Compliance		Reference
23.1327	Magnetic Direction Indicator	Yes	Deviations are within limits.	FIRG P56
23.1329	Automatic Pilot System	N/A	The Airtourer does not incorporate an automatic pilot system.	
23.1331 (a)	Instruments Using a Power Supply	Yes	A 'dry' vacuum pump is used and is monitored by a suction gauge.	
23.1331 (b)	Gyroscopic Instruments - Multi-engine Aircraft.	N/A		
23.1337 (a)	Power Plant Instruments - Instrument Lines.	Yes	Restricters fitted in critical instrument lines.	
23.1337 (b)	Power Plant Instruments - Fuel Quantity Indicator	Yes	Calibration complies with requirement.	
23.1337 (c)	Fuel Flow Meter System	N/A	No flow meter is fitted in the Airtourer.	

F.A.R. PART 23 SUB-PART F

Equipment

Part 23 Sub-Para No.	Requirement	Compliance		Reference
23.1337 (d)	Oil Quantity Indicator	Yes	A 'stick' gauge is provided for ground inspection and as no reserve tank is installed, no gauge is provided for use in flight.	
23.1351 (a)	Power Sources Adequate	Yes	Electrical Load analysis provided by AESL.	AESL Electrical Load Analysis for T6-24 Drg No.90099.
(b)	Power Source Installation & Connection	Yes	Protection, freedom from hazards and independence complied with.	TIR P28
(c)	Generator Installation	Yes	Generator can provide rated power. Regulation and cutout devices incorporated.	TIR P28
23.1353 (a)(b)	Battery Installation & Design	Yes	Battery is approved for a/c use. Installation O.K. (Aerobatic ?)	TIR P28
23.1357 (a)(b)(c)(d)	Circuit Protection	Yes	Manual reset circuit breakers are used.	Drg 90098 FTRG P43
23.1361 (a)(b)(c)	Master Switch Arrangement	Yes	Master battery switch and alternator field, switch.	Drg 90098 FTRG P43
23.1365 (a)(b)	Electric Cables	Yes	Of adequate capacity and approved type.	Drg 90098 FTRG P43

F.A.R. PART 23 SUB-PART F

Equipment

Part 23 Sub-Para No.	Requirement	Compliance		Reference
23.1367 (a)(b)(c)(d)	Switches	Yes	Switches are of an approved type.	
23.1381 (a)(b)(c)	Instrument Lights	Yes	Adequate, shielded, insulated.	DRG 90098 FTRG P43
23.1383 (a)(b)	Landing Light	Yes	With adjustment of beam height and shielding.	FTRG P43
23.1385 through 23.1401	Position & Anti-Collision Lights	Yes	Compliance has not been physically demonstrated for the Navigation lights. However the aircraft has been satisfactorily operated over a period of more than 10 years with the present system. Flashing beacon is a Grimes approved aircraft design.	
23.1411	Safety Equipment	Yes	The only safety equipment - an axe and a fire extinguisher - are readily accessible.	
23.1413	Safety Belts & Harnesses	Yes	These meet the A.N.O. requirements which cover the F.A.R. Part 23 requirements.	
23.1415	Ditching Equipment	N/A	No special ditching equipment is incorporated.	
23.1419	De-icers	N/A	No pneumatic de-icers are installed.	

F.A.R. PART 23 SUB-PART F

Equipment

Part 23 Sub-Para No.	Requirement	Compliance		Reference
23.1431	Electronic Equipment	Yes		
23.1435 (a)	Hydraulic Systems - Design	Yes	The only hydraulic system is the brake which is designed to withstand the structural and hydraulic loads.	
23.1435 (b)	Hydraulic Systems - Tests	Yes	Brake hydraulic system tested to ultimate load. Limit load is equivalent to a 100 lb pull on the brake handle.	Tested 13-10-70 witnessed M. Collins (MOT/CAD)
23.1435 (c)	Hydraulic Systems - Accumulators	Yes	The brake reservoir is on the engine side of the firewall but is unpressurised.	
23.1437	Accessories for Multi-engine Aircraft	N/A		
23.1441 23.1443 23.1447 23.1449	Oxygen Equipment & Supply	N/A	No oxygen facilities are provided on the Airtourer.	

F.A.R. PART 23 SUB-PART G

Operating Limitations & Information

Part 23 Sub-Para No.	Requirement	Compliance		Reference
23.1501	Operating Limitations & Information - Availability to Pilot & other Appropriate Crew Members.	Yes	Available in flight manual and on placards.	
23.1505 to 23.1513	Air Speed Limitations - Definitions.	Yes	Noted	
23.1519	Weight & Centre of Gravity	Yes	These are established under 23.23 and listed in the Flight Manual.	
23.1521	Power Plant Limitations	Yes	The power plant limitations are those listed for the certified engine and propeller.	
23.1523	Minimum Flight Crew	Yes	The minimum flight crew established under VFR is one pilot.	
23.1525	Kinds of Operation	Yes		
23.1527	Maximum Operating Altitude	N/A	The Airtourer is not pressurised, turbine powered or turbo-supercharged.	
23.1529	Maintenance Manual	Yes	AESL have supplied a satisfactory maintenance manual.	

F.A.R. PART 23 SUB-PART G

Operating Limitations & Information

Part 23 Sub-Para No.	Requirement	Compliance		Reference
23.1541 through 23.1567	Markings & Placards	Yes	Marking and placards meet the requirement	
23.1581 - 23.1589	Flight Manual	Yes	Flight manual and placards together meet the requirement for supply of information.	

F.A.R. PART 23 SUB-PART

APPENDIX A

Part 23 Sub-Para No.	Requirement	Compliance		Reference
A23.1 (a)	Equivalent of 23.321-23.459	Yes	Single engine aircraft with max. A.U.W. less than 6,000 lbs.	
A23.1 (b)	Symbols & Nomenclature	Yes		
A23.3	Special Symbols	Yes		
A23.5	Certif. in more than 1 Category	Yes	Wings, control surfaces investigated at max. "N1W" (Aerobatic Category). Special unsymmetrical flight load cases also investigated for acrobatic category.	REPORTS ANZ AT-001 & ANZ AT-010
A23.7 (a)	Flight Loads - Independent of Altitude. Only consider max. Design Weight.	Yes	Loads considered at I.S.A. Sea Level. Max. design weight condition considered.	REPORT ANZ AT-001
A23.7 (b)	Use of Tables & Figs to Determine $n_1 \rightarrow n_4$.	Yes	Table 1, and Figs 3 & 4 were used to determine $n_1 - n_4$ in the three categories.	REPORT ANZ AT-001
A23.7 (c)	n_3 & n_4 at min. Weight.	Yes	Figs 1 & 2 were used to determine n_3 & n_4 at the min. flying weight in each category.	REPORT ANZ AT-001

F.A.R. PART 23 SUB-PART

APPENDIX A

Part 23 Sub-Para No.		Requirement	Compliance		Reference
A23.7 (d)		Selection & Substantiation of c.g. Range.	Yes	A c.g. range has been selected and the fuselage structure has been investigated for the most adverse weight configurations.	REPORT ANZ AT-001
A23.7 (e)	1.	Equilibrium	Yes	Magnitude of aerodynamic wing loads taken as 1.05 times aircraft normal loads, in direction normal to the relative wind, for positive flight conditions. (1.0 x for negative conditions). Each component of wing loads considered.	REPORT ANZ AT-001
	2.	Min. Design Air Speeds	Yes	Min. design speeds used are not less than those specified in Fig.3 of Appendix 'A' (Feb.1965).	REPORT ANZ AT-001
	3.	Flight Load Factor	Yes	Table 1 understood and used.	REPORT ANZ AT-001
A23.9 (a)		Flight Conditions - General	Yes	Each condition of (b) and (c) investigated using a V-n diagram similar to Fig.4.	REPORT ANZ AT-001
A23.9 (b)	1.	Symmetrical conditions - Conditions of Fig.4	Yes Yes	Aircraft is designed for conditions 'A','D','E' & 'G' as in Fig.4. For 'D' & 'E', load factors and speeds comply with Fig.4, Table 1 and Fig.3 of Appendix 'A'. For 'A' & 'G', load factors correspond to Table 1 and design speeds are computed with max. static lift coefficient. Conditions 'C' & 'F' not investigated since $n_3\frac{W}{S}$ and $n_4\frac{W}{S}$ are less than $n_1\frac{W}{S}$ and $n_2\frac{W}{S}$ respectively.	REPORT ANZ AT-001

F.A.R. PART 23 SUB-PART

APPENDIX A

Part 23 Sub-Para No.		Requirement	Compliance		Reference
A23.9 (b)	2.	Symmetrical Conditions Design for Flaps etc.	Yes	Aircraft designed for two flap down conditions of Table 1. The speed used = $V_{F\,min}$ from Fig.3.	REPORT ANZ AT-001
A23.9 (c)	1.	Unsymmetrical Conditions Aft Fuselage-Wing Attachment	Yes		
	2.	Wing & Wing Carry-through	Yes	The wing and wing carry-through structures have been designed for 100% condition 'A' loading on one side and 60% on the opposite side, in the acrobatic category.	REPORT ANZ AT-001
	3.	Aileron Displacement Torsion	Yes	The combination of 75% positive manoeuvering wing loading on both sides and max. wing torsion due to aileron displacement was investigated. The resulting loads were such as to be covered by the IA, and ID + aileron defl. conditions.	REPORT ANZ AT-001
	4.	Estimation of critical.	Yes	Critical was determined by the method of A23.9(c)4.	REPORT ANZ AT-001
A23.9 (d)	1.	Supplementary Conditions Rear Lift Truss	N/A	Aircraft has no rear lift truss - fully cantilevered wing.	
	2.	Engine Mount - Torque	Yes	Covered by structural test	REPORT ANZ AT-002
	3.	Engine Mount - Lateral Loads	Yes	Covered by analysis	REPORT ANZ AT-002

APPENDIX A

Part 23 Sub-Para No.	Requirement	Compliance		Reference
A23.11 (a)	Control Surface Loads General	Yes	Control surface loads evaluated in accordance with paragraphs (b) and (c) of this section.	
A23.11 (b)	Limit Pilot Forces	Yes	Where required, control surface airloads have been reduced to correspond to the max. limit pilot forces specified in 23.397(b). Tabs were not considered since these can only be set on the ground.	REPORT ANZ AT-003 REPORT ANZ AT-005
A23.11 (c) 1.	Surface Loading Conditions Limit Loadings & Distributions	Yes	The simplified loadings given in Table 2 and Figs 5 & 6 have been used for design of the horizontal tail, vertical tail, aileron and wing flaps. Note; Wing flaps tested to aileron distribution	REPORT ANZ AT-003 REPORT ANZ AT-005
2.	Unsymmetrical Load for Acrobatic Category.	Yes	For the acrobatic category, the horizontal tail has been investigated for the unsymmetrical load specified in this paragraph. This loading was critical for the fuselage only.	REPORT ANZ AT-003
A23.11 (d)	Outboard Fins	N/A	The Airtourer has no outboard fins.	
A23.11 (e)	Special Devices	N/A	The Airtourer has no special devices.	

F.A.R. PART 23 SUB-PART

APPENDIX A

Part 23 Sub-Para No.	Requirement	Compliance		Reference
A23.13 (a)	Control System Loads Primary Flight Controls & Systems	Yes	The elevator, aileron and rudder controls were tested for the maximum pilot efforst as per 23.397(b).	STRUCTURAL TEST REPORT ANZ AT-006
A23.13 (b)	Dual Controls	Yes	This requirement has been satisfied by structural testing.	STRUCTURAL TEST REPORTS ANZ AT-006 & AT-008
A23.13 (c)	Ground Gust Conditions	Yes	As pilot forces greater than the minimum have been used in designing the control system, no investigation of ground gusts is necessary. (23.415(a)(2))	
A23.13 (d)	Secondary Controls & Systems	Yes	The brakes and tab controls have been designed for the maximum forces that a pilot is likely to apply to those controls.	

HEAD OFFICE: Airways House, Customs Street East, Auckland, New Zealand.
Telegrams & Cables: "Enzedair" Telex NZ2541.
Postal Address: Private Bag, Auckland, 1. Telephone 78-919. After Hours 78-445.

HE:T:953/10A

9 November, 1970

Mr H.W. Robertson,
Engineering Manager,
Aero Engine Services Ltd,
R.D.2,
HAMILTON.

Dear Sir,

Enclosed for your information please find the final copy of Air New Zealand Report ANZ AT-013 - "Aero Engine Services Ltd T-6 Airtourer : Compliance Check List".

Yours faithfully,
AIR NEW ZEALAND LIMITED

(J.C.M. Frost)
TECHNICAL SERVICES ENGINEER

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Robinson

for (J.C.M. Frost)
TECHNICAL SERVICES ENGINEER